Minor Project

Static Website Hosting Using EC2 and S3 (Nginx)

1. Project Objective

Launch an EC2 instance, install the Nginx web server, create an S3 bucket, upload website backup files to S3, and access the website using the EC2 public IP.

Architecture

User → EC2 Instance (Nginx Web Server) → S3 Bucket (Backup Storage)

Configuration Summary

Item	Value
Operating system	Amazon Linux 2023
Instance type	Used t3.micro (t2.micro is not the free-tier option in my region)
Web server	Nginx
Security group rules	SSH (22) from My IP; HTTP (80) from Anywhere (0.0.0.0/0)
IAM role	Role with S3 access attached to the instance
Storage	Private S3 bucket (Block all public access ON)

2. Step 1: Create the IAM Role

EC2 needs permission to write to S3. Create the role first so it can be attached at launch.

1. Open the AWS Console and go to **IAM** → **Roles** → **Create role**.
2. Trusted entity: **AWS service**. Use case: **EC2**. Click **Next**.
3. Role name: `EC2-S3-Access`.
4. Default policies (SSM and CloudWatch) can stay; they are harmless.
5. Additional policy: choose **Use existing policy**, search for **AmazonS3FullAccess**, and tick it. Do not pick the ReadOnly version, because uploads need write access.
6. Click **Create role**.

3. Step 2: Launch the EC2 Instance

Go to **EC2** → **Launch instance** and configure the following.

3.1 Name and Image

1. Name: `nginx-web-server`.
2. Application and OS Image: **Amazon Linux 2023 AMI**.

3. Instance type: **t3.micro** (free-tier eligible).

3.2 Key Pair

1. Click **Create new key pair**. Name it (for example `nginx-key`), type **RSA**, format **.pem**.
2. Save the downloaded file safely. It cannot be downloaded again.

3.3 Network Settings (Security Group)

1. Network: default VPC. Subnet: No preference. **Auto-assign public IP: Enable**.
2. Firewall: **Create security group**.
3. Tick **Allow SSH traffic from** and change the source from Anywhere to **My IP**.
4. Tick **Allow HTTP traffic from the internet** (port 80).
5. Leave **Allow HTTPS traffic** unticked. HTTPS would need an SSL certificate, which this project does not use.

Type	Port	Source	Purpose
SSH	22	My IP	Remote login to the server
HTTP	80	0.0.0.0/0	Public access to the website

3.4 Advanced Details

1. Scroll to **File systems** and select **None**. No file system is mounted in this project.
2. In **Advanced details**, open **IAM instance profile** and select the role created in Step 0. If it is missing, click the refresh icon beside **Create new IAM profile**.
3. Leave all other Advanced details settings at their defaults.

3.5 Launch

1. Review the summary, then click **Launch instance**.
2. Go to **EC2 → Instances** and wait for **Instance state: Running** and **Status check: 2/2 checks passed**.
3. Select the instance and copy its **Public IPv4 address**.

4. Step 3: Connect to the Instance

Option A: EC2 Instance Connect (browser, easiest)

1. Select the instance and click **Connect**.
2. Choose the **EC2 Instance Connect** tab, keep username `ec2-user`, and click **Connect**.

Option B: SSH from your computer

```
chmod 400 nginx-key.pem
ssh -i nginx-key.pem ec2-user@<EC2-PUBLIC-IP>
```

5. Step 4: Install and Start Nginx

Run the commands one at a time:

```
sudo dnf update -y
sudo dnf install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
sudo systemctl status nginx
curl localhost
```

Press `q` to leave the status view. Then open **http://<EC2-PUBLIC-IP>** in a browser to see the default Nginx welcome page.

6. Step 5: Create the Website

1. Open the file: `sudo nano /usr/share/nginx/html/index.html`
2. Write the sample page with a single command:

```
echo '<h1>Hello from my AWS EC2 Nginx Server!</h1>' | sudo tee
/usr/share/nginx/html/index.html
```

3. Verify on the server. The command should print the new heading, not the Welcome to nginx page:

```
curl localhost
```

4. Open **http://<EC2-PUBLIC-IP>** in a browser and confirm the new page is shown.

7. Step 5: Create the S3 Bucket and Upload the Backup

7.1 Create the bucket

1. Go to **S3** → **Create bucket**.
2. Bucket name: a globally unique name, for example `arpan-nginx-backup-2026`.
3. Region: the same region as the EC2 instance.
4. Keep **Block all public access** turned ON. Backups should stay private.
5. Click **Create bucket**.

7.2 Verify the instance can reach S3

```
aws sts get-caller-identity
```

If this prints an account and role, the IAM role is attached. If it errors, go to **EC2** → **Actions** → **Security** → **Modify IAM role** and attach the role.

7.3 Upload the backup

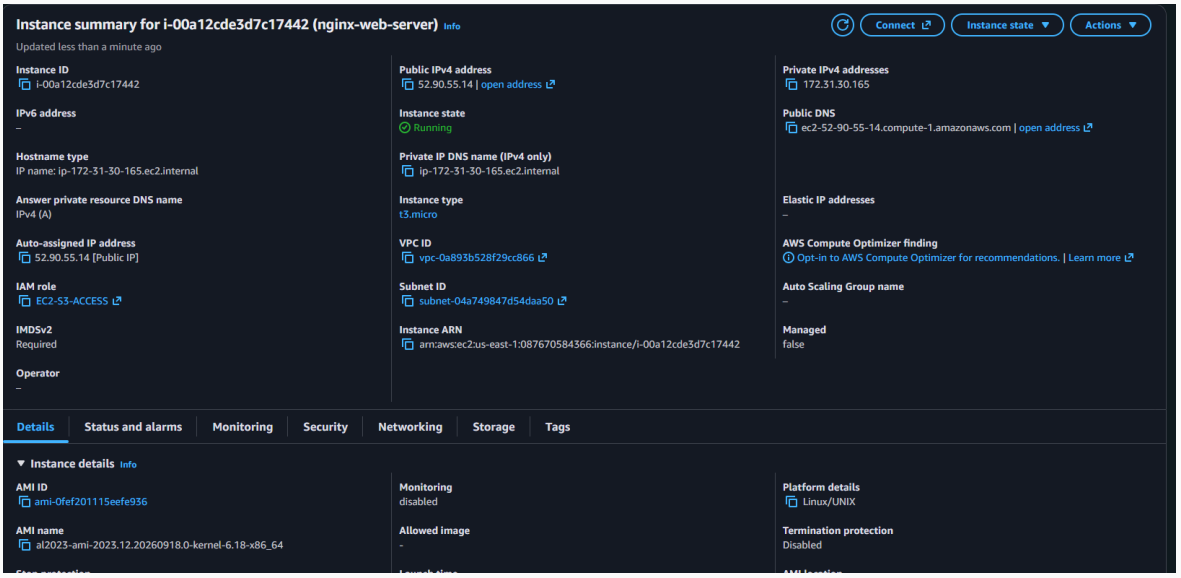
Replace the bucket name with yours. Use either option (or both):

```
aws s3 sync /usr/share/nginx/html
s3://arpan-nginx-backup-2026/website/
aws s3 ls s3://arpan-nginx-backup-2026/ --recursive
```

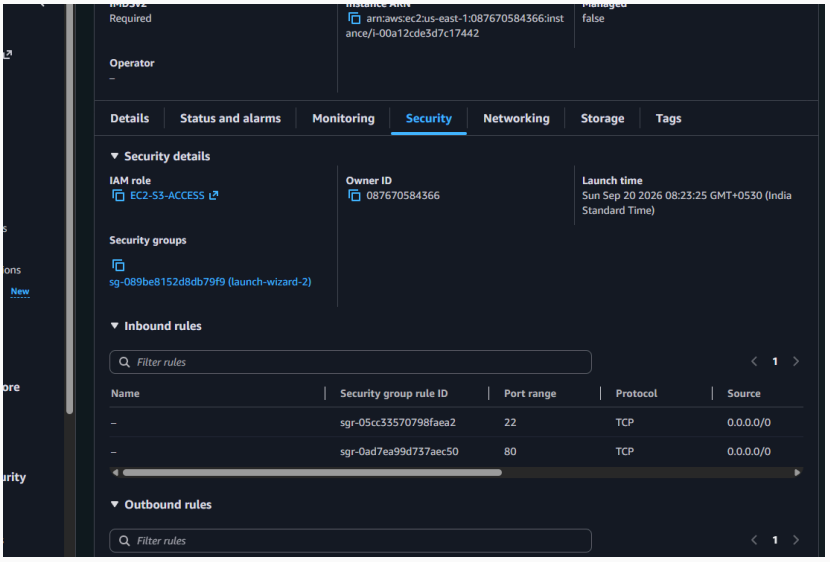
8. Deliverables: Screenshots

Capture the following and paste each into the placeholder below.

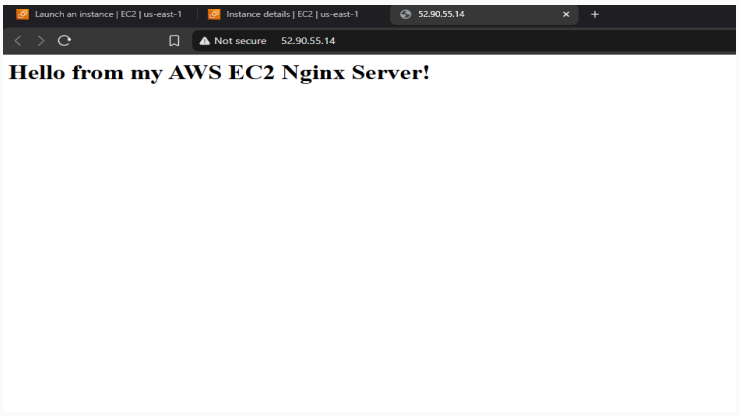
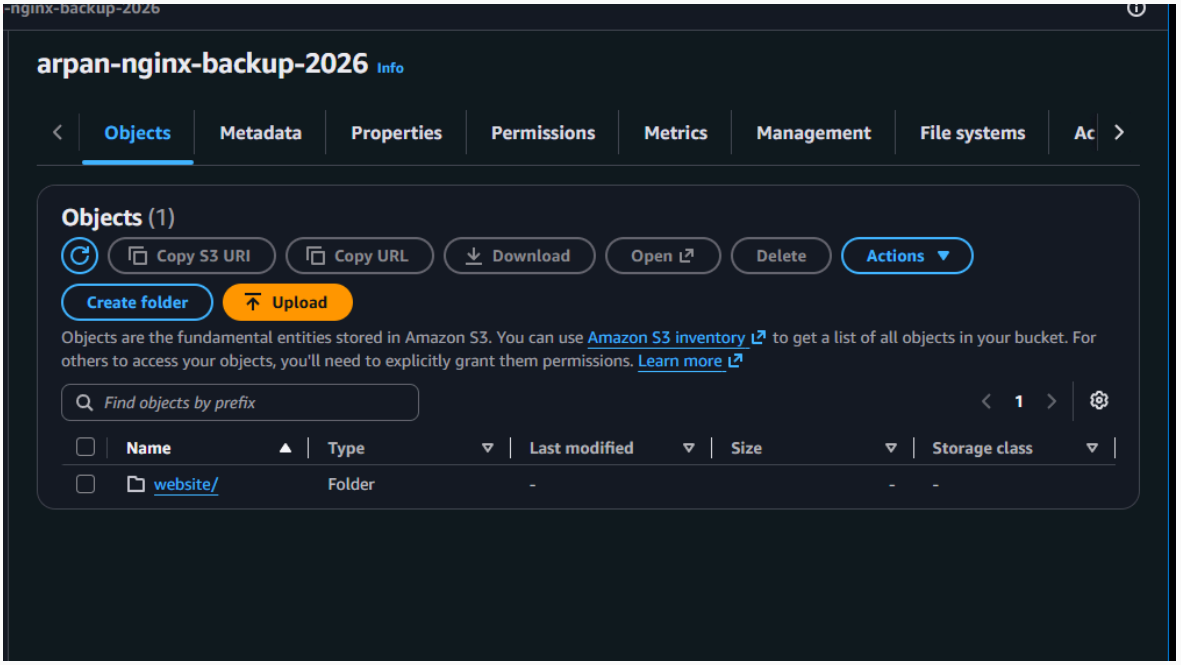
#	Screenshot	What to show
1	**EC2 instance**	Instances list showing the instance Running with its public IP

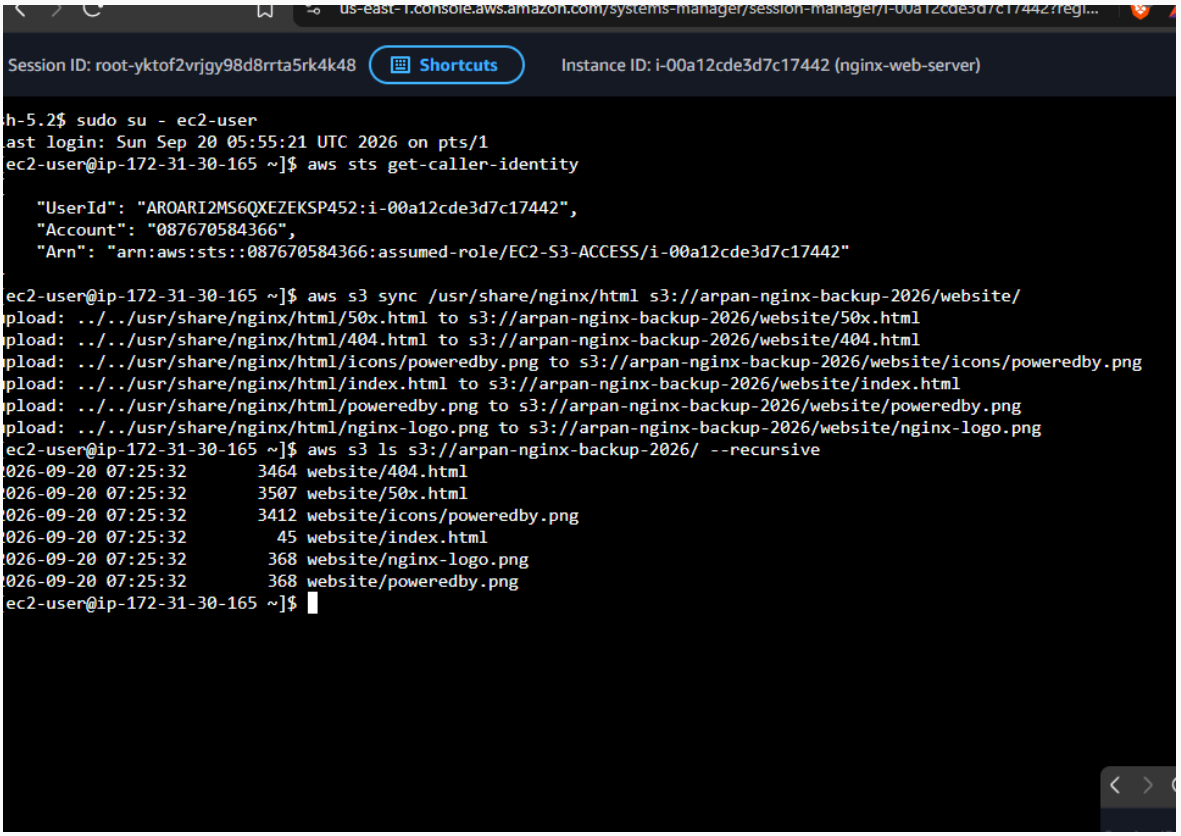


2	**Security group**	Inbound rules showing ports 22 and 80

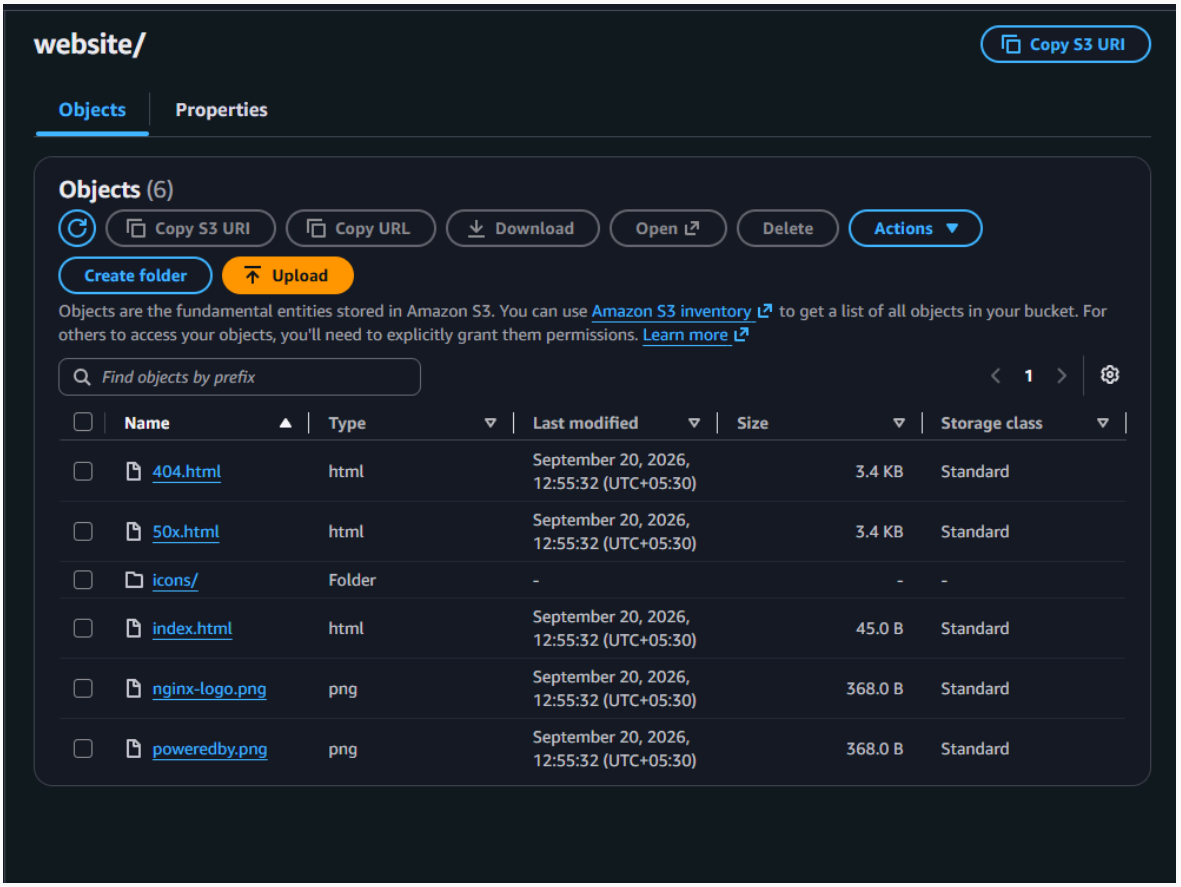


3	**Nginx website**	Browser showing your page, with the IP visible in the address bar

#	Screenshot	What to show
		
4	**S3 bucket**	Bucket overview page
		
5	**Uploaded files**	Objects inside the bucket


Session ID: root-yktof2vrjgy98d8rrta5rk4k48
Shortcuts
Instance ID: i-00a12cde3d7c17442 (nginx-web-server)
h-5.2$ sudo su - ec2-user
ast login: Sun Sep 20 05:55:21 UTC 2026 on pts/1
ec2-user@ip-172-31-30-165 ~]$ aws sts get-caller-identity
"UserId": "AROARI2MS6QXEZEKSP452:i-00a12cde3d7c17442",
"Account": "087670584366",
"Arn": "arn:aws:sts::087670584366:assumed-role/EC2-S3-ACCESS/i-00a12cde3d7c17442"
ec2-user@ip-172-31-30-165 ~]$ aws s3 sync /usr/share/nginx/html s3://arpan-nginx-backup-2026/website/
pload: ../../usr/share/nginx/html/50x.html to s3://arpan-nginx-backup-2026/website/50x.html
pload: ../../usr/share/nginx/html/404.html to s3://arpan-nginx-backup-2026/website/404.html
pload: ../../usr/share/nginx/html/icons/poweredby.png to s3://arpan-nginx-backup-2026/website/icons/poweredby.png
pload: ../../usr/share/nginx/html/index.html to s3://arpan-nginx-backup-2026/website/index.html
pload: ../../usr/share/nginx/html/poweredby.png to s3://arpan-nginx-backup-2026/website/poweredby.png
pload: ../../usr/share/nginx/html/nginx-logo.png to s3://arpan-nginx-backup-2026/website/nginx-logo.png
ec2-user@ip-172-31-30-165 ~]$ aws s3 ls s3://arpan-nginx-backup-2026/ --recursive
026-09-20 07:25:32 3464 website/404.html
026-09-20 07:25:32 3507 website/50x.html
026-09-20 07:25:32 3412 website/icons/poweredby.png
026-09-20 07:25:32 45 website/index.html
026-09-20 07:25:32 368 website/nginx-logo.png
026-09-20 07:25:32 368 website/poweredby.png
ec2-user@ip-172-31-30-165 ~]$

#	Screenshot	What to show



9. Troubleshooting

Problem	Fix
Website does not load	Check port 80 is open in the security group, use http:// not https://, and confirm Nginx is running.
SSH connection times out	Your IP changed. Update the port 22 rule to My IP, or use EC2 Instance Connect.
aws s3: Unable to locate credentials / Access Denied	The IAM role is not attached. EC2 → Actions → Security → Modify IAM role.
IAM role missing in the launch dropdown	Click the refresh icon next to Create new IAM profile, or create the role first.
Still seeing the Nginx welcome page	index.html was not saved to /usr/share/nginx/html/. Re-check the path and hard-refresh the browser.

10. Learning Outcomes

- EC2 instance launch and security group configuration

- Connecting to a Linux server over SSH
- Installing and managing Nginx as a web server
- Hosting a static website on a public IP
- Using S3 for backup storage and IAM roles for secure access